Noise bends a polynomial fit

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Abstract

We measure how noise bends a polynomial fit, and find that its error grows as the square of the noise.

Introduction

State of the art

Polynomial fits are standard [Smith and Li, 2020, Jones, 2019]. Noise is known to bias them [Lee and Hall, 2018]. An earlier draft said so. A lost work claimed it first [to be resolved], and a late one agreed [Late, 2024]. A third said it again.

Results

How does noise change the fit?

The error grows with the noise

The error of the fit follows

$$e(\sigma) = 0.8\,\sigma^2 + 0.02 \tag{1}$$

and Eq. 1 holds at every noise level we ran, as Fig. 1 shows. Table 1 lists the errors. The fit follows Smith's method [Smith and Li, 2020].

Table 1: The error of the fit at each noise level.

noise	rmse
0.1	0.03
0.2	0.05
0.4	0.15

Figure 1: The error of the fit grows with the noise.

Discussion

Limits

The noise model is Gaussian, as Lee assumed [Lee and Hall, 2018]. A heavier tail would bend the fit further, and the next runs should try one.

Methods

Fitting

We fit by least squares, as Jones did [Jones, 2019].

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Competing interests

The authors declare none.

Data availability

The data are synthetic and the project's runs made them.

Code availability

[TO BE GIVEN: the code repository, which setup has not settled]

References

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